

*Research Article***Pre and Postnatal Exposure of Baicalein (Flavonoid) on Developmental Landmarks of Mice****Vaadala Sridevi, Mythatha Grace Sugandha Sowjanya, Pamuru Ramachandra Reddy***

Department of Biochemistry, Yogi Vemana University, Vemanapuram, Kadapa – 516 003, A.P. India

Abstract**Background:** Baicalein, a flavonoid comes under phytoestrogen family found to have estrogenic activity in mammals. The present investigation aimed to study the effect of baicalein on behaviour and developmental land marks of Wistar mice.**Methods:** Baicalein at a dose of 30, 60 and 90 mg/Kg body weight were injected to three different groups of pregnant female mice on gestation days 11, 13, 15 and 17, whereas controls were injected with vehicle. Controls and experimental groups were allowed to deliver pups and baicalein exposure was continued during lactation.**Results:** No clinical signs of toxicity were observed in the behaviour of experimental animals during experimentation when compared to controls. Also significant decrease ($P<0.05$) in number of live pups in baicalein received females observed when compared to controls. Also observed decrease in the survival index of pups on postnatal day 4 and 21 in experimental groups. Significant ($P<0.05$) decrease in early onset of puberty in females (vaginal opening) and delay in testes descent in males of experimental groups were also observed. Also observed significant decrease ($P<0.05$) in the body weight of pups exposed perinatally to baicalein on postnatal day (PND) 1, 7, 14 and 21, along with significant decrease ($P<0.05$) in crown rump length when compared to controls. No significant change was observed in pinna detachment, eye opening and fur development in experimental groups when compared to controls.**Conclusion:** It was concluded that baicalein interferes in developmental landmarks in mice.**Keywords:** Baicalein, Perinatal, Developmental landmarks, Puberty, Testes descent**1. Introduction**

Studies on animals provide an evidence of phytoestrogens on developmental land marks and behavioural studies. Due to estrogenic activity of phytoestrogens, they may cause the hormonal effects which might be harmful to the developing embryo when consumed during the pregnancy and lactation. It has been recognized that the phytoestrogens interact with the mammalian endocrine system (1-3) and causes endocrine disruption. Hence these phytoestrogens are called as endocrine disrupting chemicals (EDCs). Exposure of these chemicals in humans affects during development and adulthood (4, 5). The EDCs are exposed by the embryos during early stages of life, if the pregnant mother is exposed during and/or after pregnancy. As these chemicals are lipophilic they dissolve easily in fat, stored in the body fat and they are used as a fuel for the development of the embryo and also useful for the Production of milk during lactation and hence the offspring's are exposed pre- and postnatally to these chemicals (6). EDCs including phytoestrogens show adverse effects on organism by altering the endocrine functions (7-9). The action of these EDCs is mainly on endocrine

and reproductive systems by binding to the nuclear receptors, non-nuclear steroid hormone receptors i.e., membrane estrogen receptors (8). Besides this phytoestrogens are also cross the placental barrier and reaches to developing embryo (10). The pregnant women exposed to phytoestrogens alone they show their effects on the development of the embryo. Baicalein is flavonoid phytoestrogen and is commonly called as skull cap and is used to treat uterus and ovary inflammation. Like other phytoestrogens baicalein also mimicks like an estrogen and binds to the estrogen receptors there by disrupting the endocrine system in vertebrates (11). In spite of this, studies need to be focussed on parameter of the developmental landmarks like body weight, fur development, survival index, eye opening, pinna detachment, testes descent, vaginal opening and puberty gain including number of live pups. Very few reports explained the role of selected phytoestrogens and their effects on the parameters of development and behaviour of young rats (12, 13). Conversely the literature is having lacuna on developmental land marks and behavioural studies in animals by the exposure of phytoestrogens and their effects. However the phytoestrogen induced toxicity at the development of embryo and its growth is still in its infancy. Furthermore studies are need to know the toxicity caused by phytoestrogens

***Corresponding address:** Dr Pamuru Ramachandra Reddy, Department of Biochemistry, Yogi Vemana University, Vemanapuram, Kadapa – 516 003, A.P., India. E-mail: redidyprbiotech@gmail.com

in mammalian system and furthermore to elucidate the molecular mechanistic action of phytoestrogens on development of embryo and growth of young ones. The present investigation initiated to emphasise the role of flavonoid phytoestrogen baicalein (which is commonly used as anti-inflammatory and anti-cancer therapeutic drug for humans and animals) on behaviour and developmental landmarks of Wistar mice exposed perinatally.

2. Materials and methods

2.1. Animals and maintenance

The experimental animal model selected for the present study is the Wistar mice of age 45-50 days with an average weight of 29 ± 2 g. The experimental animals were purchased from Sri Venkateshwara Enterprises, Bangalore, Karnataka, India. Animals were maintained in a clean, well ventilated and air-conditioned room (12h:12h light:dark cycle) at $25 \pm 2^\circ\text{C}$ with a relative humidity of $50 \pm 5\%$. Sufficient amount of food and water provided every day. The mice were acclimatized for a week and then used for experimentation. The experiments were carried out in accordance with the guidelines of the Committee for the Purpose of Control and Supervision on Experiments on Animals, Government of India (CPCSEA) (14). This study was also carried out according to the guidelines for the care and use of laboratory animals (NRC, 1996) and approved by the Institutional Animal Ethical Committee at Sri Venkateswara University, Tirupati, India (Resolution No: 10/(i)/a/CPCSEA/IAEC/SVU/PSR/NG/Dt. 18-10-2010).

2.2. Test Chemicals

Commercially purchased baicalein from Sigma Aldrich, USA was used as test chemical for the present investigation. Baicalein (5, 6, 7-trihydroxyflavone) is a flavonoid derived from a common class of phytoestrogens, isolated from the roots of *Scutellaria baicalensis*. Molecular weight of baicalein is 270.241 g/mol, the chemical formula is $\text{C}_{15}\text{H}_{10}\text{O}_5$ (Figure 1) and is soluble in 100% dimethyl sulfoxide (DMSO).

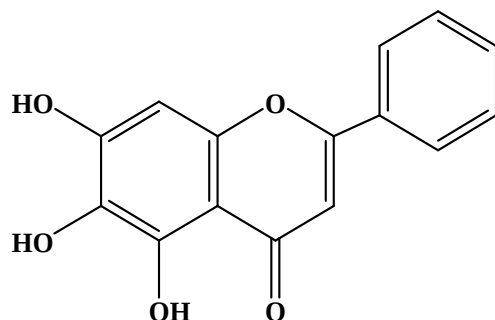


Figure 1. The chemical structure of baicalein.

2.2.1. Vaginal Smear Test

Vaginal smear cytology is the most practical method to check the estrous cycle normality. The vaginal smear was examined by the method described by Zarrow et al. (15) and reviewed by Cooper et al. (16). A few drops of saline (0.9% NaCl) were introduced carefully into the vagina by the help of pasture pipette. The saline was sucked quickly back into the pipette by releasing the pressure. The smear was carefully taken on a clean plain glass slide as a thin layer and observed under a microscope (Axiostar plus, Carl Zeiss, Germany) for various stages of estrous cycle. Based on the type of cells present in the vaginal smear, the stages were easily distinguished and determined as pro-estrus (contains epithelial cells), estrus (cornified cells), met-estrus (cornified and leukocytes present in large numbers) and di-estrus (contains leukocytes). On the evening of pro-estrus stage observed, the females were kept for cohabitation with males in separate cages.

2.2.2. Experimental design

The following day of cohabitation, the female mice were checked for vaginal plug in the early morning (before 7.0 am). The animal with vaginal plug was considered as day one of pregnancy and the pregnant females were separated from the males and kept in separate cages. Sixteen pregnant mice were divided into four groups of each with equal number. Graded doses of 30, 60, 90 mg/Kg body weight of baicalein were injected on 11th, 13th, 15th and 17th day of gestation to group 1, 2 and 3 respectively. The group 4 served as control and received 100% DMSO (Merck, Mumbai, India). All the animals were fed with normal pellet diet and water ad libitum.

The pregnant mice were allowed to deliver pups. The F1 progeny is observed for number of live pups/mice on postnatal day 1 (PND 1) and the sex ratio of male and female pups, abnormalities in the developmental landmarks. Besides this the clinical signs of toxicity in the young ones of both experimental and control mice were also recorded during and after weaning.

2.2.3. Developmental Land Marks

The accurate body weights (g) of the young ones were measured by using electronic balance on 1st, 7th, 14th and 21st day of weaning period. The number of live pups/mice is counted immediately after delivery and the sex ratio of male and female pups were counted after the weaning period. Crown rump length is the measurement of the young ones from the top of the head (crown) to the bottom of the buttocks (rump). It is measured in the natural state with an unstretched body which is usually in c-shaped. Crown rump length of young ones were measured on PND 1 immediately after delivery by

using vernier callipers. Survival index was measured on PND 1 and PND 21 and was calculated using the formula: [(Number of live off springs at lactation day 4 or 21/Number of live offspring delivered) X 100]. The other developmental land marks such as pinna unfolding, fur development, ear opening, eye opening, vaginal opening and testes descent are observed with the naked eye.

2.2.4. Statistical analysis

The data were statistically represented with two tailed ANOVA by using the software GraphPad.Prism. v5.0.3.477. The statistical significance was considered at $P < 0.05$. Results are expressed as mean $\pm$ SEM.

3. Results

In the present study, behaviour of the pregnant female mice, lactating mothers and its offspring were observed time to time and found no behavioural changes exposed to baicalein at 30, 60 and 90 mg/Kg body weight. The ratio of male and female pups was measured in experimental and control groups and found no significant change (Table 1). The number of live pups delivered by baicalein exposed and control females were measured. Significant decrease ($P < 0.05$) was observed in number of live pups delivered by baicalein exposed females when compared to the control group and the decrease in experimental groups is in a dose dependent manner (Figure. 2). On PND 1 observed significant decrease ($P < 0.05$) in crown rump length in both male and female mice pups transplacentally exposed to 30, 60 and 90 mg baicalein/Kg body weight in a dose dependent manner than the controls. Survival index of control and experimental offspring was measured on PND 4 and 21 in the present study and found reduction in survival index of 60 and 90 mg baicalein/Kg body weight exposed young ones on PND 4, whereas on PND 21 decreased survival index was recorded in all experimental young ones when compared to controls (Table 1).

The significant ($P < 0.05$) reduction in the number of days for opening of vagina was observed in perinatally 60 mg and 90 mg baicalein/Kg body weight exposed young females when compared to control and 30 mg baicalein/Kg body weight exposed young ones (Table 2). In the present study also observed no significant change in pinna detachment, fur development and eye opening (Table 2) in experimental young ones when compared to control mice. Increased number of days for testes descent was recorded and significantly increased ($P < 0.001$) in the experimental groups when compared to control mice. The increase in days for testes descent is also in dose dependent manner in baicalein exposed young mice (Table 2). Observed significant reduction ($P < 0.05$) in the body weights of offspring exposed perinatally to baicalein

(30, 60 and 90 mg/Kg body weight) on PND 1, 7, 14 and 21 in a dose dependent manner when compared to controls (Figure 3).

Table 1. Pre and postnatal exposure of baicalein on crown rump length, survival index of young mice on PND 4 and 21 and male:female ratio.

Parameter	Controls	30 mg/Kg body weight	60 mg/Kg body weight	90 mg/Kg body weight
CRL (cm) of pups on PND 1	3.85 $\pm$ 0.041	3.70* $\pm$ 0.038	3.56* $\pm$ 0.037	3.20*** $\pm$ 0.026
Survival index of pups on PND 4 (%)	46/46 (100)	40/40 (100)	34/38 (89.47)	28/33 (84.84)
Survival index of pups on PND 21 (%)	45/46 (97.82)	38/40 (95.0)	32/38 (84.21)	26/33 (78.78)
Male: Female ratio	26:20	23:17	17:21	17:16

Values are mean $\pm$ SEM of 8 young mice.

Values in parenthesis are percent change from controls.

* and *** represents significantly different from controls at P-value < 0.05 and < 0.0001 respectively.

Table 2. Effect of pre and postnatal exposure of baicalein on developmental landmarks of mice pups.

Pinna detachment	3.25 $\pm$ 0.163	3.37 $\pm$ 0.183	3.50 $\pm$ 0.189	3.62 $\pm$ 0.189
Fur development	7.37 $\pm$ 0.183	7.62 $\pm$ 0.263 (3.39)	7.75 $\pm$ 0.163 (5.15)	7.87 $\pm$ 0.226 (6.78)
Eye opening	15.75 $\pm$ 0.313	14.87 $\pm$ 0.295 (0.81)	15.25 $\pm$ 0.313 (3.38)	15.62 $\pm$ 0.183 (5.89)
Vaginal opening	34.63 $\pm$ 0.183	34.38 $\pm$ 0.183 (-0.72)	31.38*** $\pm$ 0.323 (-9.38)	28.38*** $\pm$ 0.263

Values are mean $\pm$ SEM of 8 young ones

Values in parenthesis are percent change from controls.

** and *** represents significantly different from controls at P-value < 0.001 and < 0.0001 respectively.

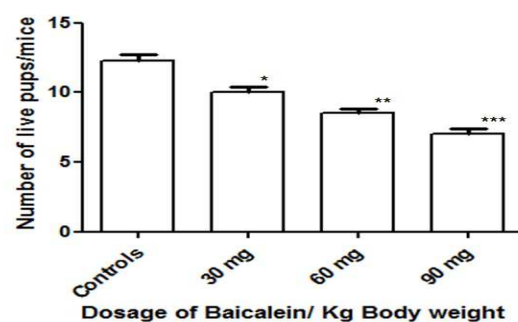


Figure 2. Tran placental effect of baicalein on number of live pups delivered by female mice.

*, ** and *** represents significantly different from controls at P-value < 0.05 , < 0.001 and < 0.0001 respectively.

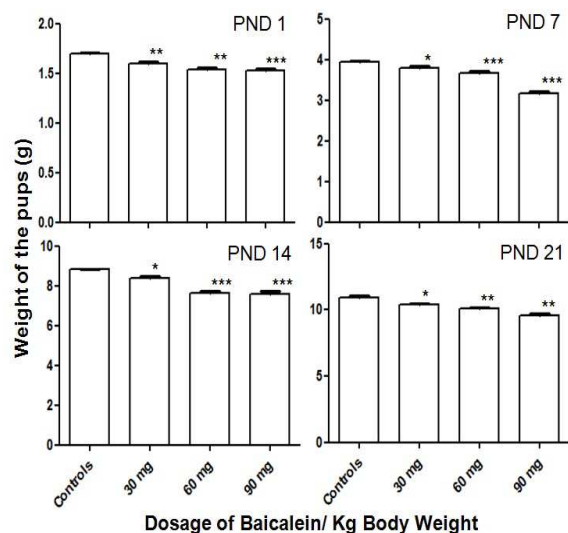


Figure 3. Perinatal effect of baicalein on body weights (g) of mice pups on postnatal days (PND) 1, 7, 14 and 21. *, ** and *** represents significantly different from controls at P-value <0.05, <0.001 and <0.0001 respectively.

4. Discussion

Phytoestrogens occurs in plant based diet and mimics like estrogens, which are predominant in curing hormonal dependent cancers and other age related diseases (17). Due to the beneficiary effects on human health, consumption of phytoestrogen based diet is increasing Worldwide. Besides this, the literature is also showing the adverse effects of phytoestrogens on developing embryo if the pregnant woman exposed it (11-13).

In the present study significant decrease ($P < 0.05$) was observed in number of live pups delivered by baicalein exposed mice when compared to the controls and decrease is in a dose dependent manner. Similar results were reported by Romero *et al.* (18) in rats exposed to 10 and 100 mg daidzein/genistein/Kg body weight during pregnancy. They observed smaller number of live litter in 10 and 100 mg daidzein/genistein exposed groups (18). Pushpalatha (19) and Harini (20) also reported decrease in the size of litter in pregnant rat and mice exposed *in utero* to proluton depot (hydroxyprogesterone) and progesterone respectively. In rats, phytoestrogen biochanin-A transplacental exposure decreased number of live pups (11, 13). Baicalein exposure to pregnant rats decreased the number of live pups in a dose dependent manner which may be due to the increased fetal toxicity of baicalein towards increased number of resorption and decreased number of implantations in a dose dependent manner. Significant decrease ($P < 0.05$) was observed in crown rump length in mice pups transplacentally exposed to baicalein in this study. Similarly Soujanya *et al.*, (13) reported the decrease in crown rump length on PND1 in rat offspring exposed transplacentally to biochanin -A(50 and 100 mg/kg

body weight). Decrease in the crown rump length on PND 1 in the young mice exposed transplacentally to baicalein in the present study may be due to altered hormone metabolism in the young mice by baicalein. Significant decrease ($P < 0.05$) was observed in the mean days for vaginal opening of female pups exposed to the baicalein transplacentally. Similar results were observed in female rodents fed with genistein or soya based diet during development showed earlier vaginal opening (21-24). Early pubertal onset has also been observed in girls exposed to EDCs pre and postnatally (25-26). Besides this irregular estrus cycles i.e. increase or decrease in cycle duration was observed in animals exposed to resveratrol (27).

In contrast to this observed significant increase in the number of days for opening vagina in rat pups exposed perinatally to biochanin-A (11, 13). The above reports supporting that the decreased days for vagina opening in baicalein exposed female young mice in this study is may be due to early in the release of pubertal hormones promoted by baicalein and this was supported by Manson and Kang (28). Reduction in the survival index of mice pups was observed on PND 4 and 21 exposed perinatally to baicalein (60 and 90 mg/Kg body weight) when compared to controls. Reports are also explained the decrease in the continued existence of young ones by phytoestrogens in rats (29). The present study results are in support with the study conducted by Sowjanya and Reddy (11) where they exposed rat offspring to isoflavone biochanin-A perinatally.

Significant decrease ($P < 0.001$) in the testes descent was observed in baicalein exposed young males when compared to controls in a dose dependent manner. The result of the present study is in correlation with the reports existing. Delayed testes descent was reported in young male rats exposed to biochanin-A perinatally (11, 13). The delay in testes descent reported in the present study is may be due to delay in the response of release of male hormones including testosterone by baicalein exposure perinatally in young male mice.

Decrease in the body weights of young mice exposed perinatally to baicalein are significant ($P < 0.05$) when compared to control mice. Similar to this, Sowjanya and Reddy (11), and Soujanya *et al.* (13) reported that the body weights of perinatally biochanin-An exposed young rat on PND 1, 7, 14 and 21 was significant decreased. Reported lower birth weights in offspring of rats exposed to genistein (30). Besides this, there is evidence that high dose estrogen diet i.e. 10% flaxseed to the pregnant rat lowers birth weights than other animals fed with other diets in offspring (31). In contrast to present result, reported genistein exposure to pregnant rats has no effect on offspring birth weights and feed intake (32-33). In the present study decrease in the body weights of offspring exposed perinatally to baicalein is may be due to suppression

of release of pituitary growth hormone by baicalein there by reduced food intake in pregnant dams and decreased appetite in offspring. This may lead to no fat depositions in dams and young ones during lactation period, ultimately results in loss of body weight of offspring during birth and lactation period.

However, no significant changes were observed in male and female sex ratio, pinna detachment, fur development and eye opening in baicalein exposed offspring when compared to controls. Similarly no change in fur development was reported in young rats exposed pre and postnatal to biochanin-A (11, 13). In the same study they also reported significant change in sex ratio of male and female pups, pinna detachment and eye opening in the rat offspring exposed perinatally to biochanin-A which was not witnessed in the present study.

Many reports are supported the results of present study. Manson and Kang (28) reported that pregnancy and lactation periods are hormone sensitive for offspring during which sexual differentiation of the fetus reproductive tract and central nervous system will take place under the influence of hormones. Similarly, Whitten *et al.* (34-35) in their study reported that phytoestrogens are transferred to the offspring through rat dam's milk and affects its reproduction at adolescence and reproductive abnormalities like disrupted estrus cycles in females and decreased sexual behaviour in male offspring. Zimmerman *et al.* (29) noticed that taking of phytoestrogens in pregnant animals showed its effect on the fetus which leads to decreased litter size, pup survival and increased resorption or abortion. Flavonoids of flax seeds showed its effects on reproductive development of offspring exposed during pregnancy or lactation (12). Humans are mainly exposed to the environmental estrogens including phytoestrogens through diet (36) which may cause adverse effects on development and reproduction and are open for researchers.

Baicalein is a phytoestrogen and comes under flavonoid family, shows estrogenic activity in animals and humans. Estrogens execute reproductive development and development of secondary sexual characters at physiological levels. But when taken in excess amounts they exert detrimental consequences on development and had its effects on the development of reproductive organs thereby decreasing the fertility of the animal. Which has been proven in the present study that the exposure of baicalein in to pregnant mice showed decrease in live litter size, decreased crown rump length and survival index in pups, early vaginal opening in female young ones and delayed testes descent in male young ones and decrease in body weights were observed. Like other phytoestrogens baicalein in moderate and high concentrations may transmit to developing embryo through the placenta

and to the young ones through milk during weaning period and may have the capacity to alter the hormone concentrations in the young ones and causes the alteration in physiology thereby developmental and reproductive abnormalities.

In conclusion, it is clear from the present study that baicalein may influence the development and growth of the young mice exposed perinatally in a dose dependent manner without showing behavioural abnormalities. However further studies are in progress to characterize the baicalein effects on mice development and reproduction in this laboratory.

Acknowledgements

Authors are thankful to Prof. P. Sreenivasula Reddy, Department of Zoology, Sri Venkateswara University, and Tirupati, India for guidance and for providing animal facility for conduction of the experiments. V. Sridevi acknowledges Department of Science and technology (Inspire Fellow, Code: IF131094) for awarding fellowship and also thankful to Mr. Viswarupaachari, M.Sc. student of Department of Biochemistry, Yogi Vemana University, Kadapa for helping in maintain the mice colony and support during the experimentation.

Conflict of interest Statement

The authors declare no conflict of interest.

References

- McCarrison R. The goitrogenic action of soya-bean and ground-nut. *Indian J Med Res* 1933;21:179-81.
- Patisaul HB, Jefferson W. The pros and cons of phytoestrogens. *Front Neuroendocrinol* 2010;31:400-19.
- Bennetts HW, Underwood EJ, Shier FL. A specific breeding problem of sheep on subterranean clover pastures in Western Australia. *Aust Vet J* 1946;22:2.
- Foster WG, Agzarian J. Toward less confusing terminology in endocrine disruptor research. *J Toxicol Environ Health B Crit Rev* 2008;11: 152-61.
- Abaci A, Demir K, Bober E, Buyukgebiz A. Endocrine disrupters - with special emphasis on sexual development. *Pediatr Endocrinol Rev* 2009;6:464-75.
- Colborn T, Frederick S, Vom Saal A, Soto. Developmental Effects of Endocrine-Disrupting Chemicals in Wildlife and Humans. *Environ Health Perspect* 1993;101:378-84.
- Lee MM. Endocrine Disrupters, a Current Review of Pediatric Endocrinology. 2007;109-18.
- Phillips KP, Foster WG. Key developments in endocrine disrupter research and human health. *J Toxicol Environ Health B Crit Rev* 2008;11: 322-44.

9. Wang MH, Baskin LS. Endocrine disruptors, genital development, and hypospadias. *J Androl* 2008;29:499-505.
11. Todaka E, Sakurai K, Fukata H, et al. Fetal exposure to phytoestrogens - the difference in phytoestrogen status between mother and fetus. *Environ Res* 2005;99:195-203.
12. Sowjanya MGS, Ramachandra Reddy P. Phytoestrogen biochanin-A alters reproductive health in rats: Perinatal exposure on reproduction. Lambert Academic Publishing, Germany. 2015.
13. Tou JC, Chen J, Thompson LU. Flaxseed and its lignan precursor, secoisolariciresinol diglycoside, affect pregnancy outcome and reproductive development in rats. *J Nutr* 1998;128:1861-8.
14. Soujanya MGS, Ramachandra Reddy P, Sreenivasula Reddy P. Perinatal exposure of biochanin-A induced abnormalities in offspring of rats. *J infertil reprod biol* 2014;2:115-23.
15. CPCSEA. CPCSEA guidelines for laboratory animal facility. *Indian J Pharmacol* 2003;35:257-74.
16. Zarrow MX, Yochim JM, McCarthy JL. *Experimental Endocrinology: A Sourcebook of Basic Techniques*, New York. 1964.
17. Cooper RL, Goldman JM, Vandenberg JG. Monitoring of the estrous cycle in the laboratory rodent by vaginal lavage. In *Methods in Toxicology: Female Reproductive Toxicology*. New York, Academic Press.1993;45-54.
18. Edwin DL, Kenneth DRS, Trent DL. Action of environmental estrogens on neural circuits and behaviour. *Phytoestrogens: hormonal action and brain plasticity*. *Brain Res Bull* 2005;65:193-8.
19. Romero V, Dela Cruz C, Pereira OCM. Reproductive and toxicological effects of isoflavones on female offspring of rats exposed during pregnancy. *Anim Reprod*. 2008; 5: 83-89.
20. Pushpalath T. In utero exposure of proluton depot on male reproduction in rats. Thesis submitted to Sri Venkateswara University, Tirupati, A.P., and India. 2004.
21. Harini C. Studies on effect of progesterone on male reproduction in mice. Thesis submitted to Sri Venkateswara University, Tirupati, A.P., India. 2007.
22. Casanova M, You L, Gaido KW, et al. Developmental effects of dietary phytoestrogens in Sprague-Dawley rats and interactions of genistein and daidzein with rat estrogen receptors alpha and beta in vitro. *Toxicological Sci* 1999;51:236-44.
23. Gallo D, Cantelmo F, Distefano M, et al. Reproductive effects of dietary soy in female Wistar rats. *Food Chem Toxicol* 1999;37:493-502.
24. Lewis RW, Brooks N, Milburn GM, et al. The effects of the phytoestrogen genistein on the postnatal development of the rat. *Toxicol Sci* 2003;71:74-83.
25. Bateman HL, Patisaul HB. Disrupted female reproductive physiology following neonatal exposure to phytoestrogens or estrogen specific ligands is associated with decreased GnRH activation and kisspeptin fiber density in the hypothalamus. *Neurotoxicol* 2008;29:988-97.
26. Blanck HM, Marcus M, Tolbert PE, et al. Age at menarche and Tanner stage in girls exposed in utero and postnatally to polybrominated bisphenyl. *Epidemiol* 2000;11:641-7.
27. Vasiliu O, Muttineni J, Karmaus W. In utero exposure to organochlorines and age at menarche. *Hum Reprod* 2004;19:1506-12.
28. Freyberger A, Hartmann E, Hildebrand H, Krottinger F. Differential response of immature rat uterine tissue to ethynylestradiol and the red wine constituent resveratrol. *Arch Toxicol* 2001;74:709-15.
29. Manson JM, Kang YJ. Test methods for assessing female reproductive and developmental toxicology. In: *Principles and Methods of Toxicology* (Hayes A. W, ed.), Ravens Press Ltd., New York, NY. 1989;2:311-59.
30. Nagao T, Yoshimura S, Saito Y, et al. Reproductive effects in male and female rats of neonatal exposure to genistein. *Reprod Toxicol* 2001;15:399-411.
31. Wiedemann E. Adrenal and gonadal steroids. In: *Endocrine Control of Growth* (Daughaday, W. H., (Ed.), Elsevier, New York, NY. 1981; 67-119.
32. Odum J, Tinwell H, Jones K, et al. Effect of rodent diets on the sexual development of the rat. *Toxicol Sci* 2001;61:115-27.
33. Zimmermann SA, Clavenger WR, Brimhall BB, Bradshaw WS. Diethylstilbestrol induced perinatal lethality in the rat. II. Perturbation of parturition. *Biol Reprod* 1991;44:583-9.
- Haumann BF. Soy protein foods gain storage space. *INFORM* 1997;6:588-96.
34. Soucy NV, Parkinson HD, Sochaski MA, et al. Kinetics of genistein and its conjugated metabolites in pregnant Sprague-dawley rats following single and repeated genistein administration. *Toxicol Sci* 2006;90:230-40.
35. Whitten PL, Lewis C, Naftolin F. A phytoestrogen diet induces the premature an ovulatory syndrome in lactationally exposed female rats. *Biol Reprod* 1993;49:1117-21.
36. Whitten PL, Lewis C, Russel E, Naftolin F. Potential adverse effects of phytoestrogens. *J Nutr* 1995;125:771s-776s.